

Analiza complexitatii pentru un algoritmul numărarea cifrelor unui număr(iterativ) CMF/CMD

```
int sumaDeasupraDiagonalei(int a[][100], int n) {
    int suma = 0;
    for (int i = 0; i < n; i++) {
        for (int j = i + 1; j < n; j++) {
            if (a[i][j] < 0) break;
            suma += a[i][j];
        }
    }
    return suma;
}
```

Instructiune	Cost	Nr. Rep
int suma = 0;	c	1
for (int i = 0; i < n; i++)	c + c(n+1) + cn	n + 1
for (int j = i + 1; j < n; j++)	c + c(n+1) + cn	n*(n + 1)
if (a[i][j] < 0)	c	$\delta(n)$
suma += a[i][j];	c	$n * (n-1)/2$
break	c	n-1
suma += a[i][j];	c	$\delta(n)$
return suma;	c	1

	c	c(n+1)	cn
for	i=0	i<n	i++
for	j=i + 1	j<n	j++

CMF: $\forall i, a[i][i+1] < 0 \Rightarrow \delta(n) = 0$

CMD: $(n) = c_1 + c_2 \cdot n + c_3 \cdot (n-1) + c_4 \cdot \delta(n)$

TCMF(n) = $c \cdot 1 + (c \cdot 1 + c(n+1) + cn) + n(c \cdot 1 + c(n+1) + cn) + c \cdot n^2 + c \cdot n(n-1)/2 + c \cdot 1 \rightarrow O(n)$

TCMD(n) = $\rightarrow O(n)$